BATCH & BUILD · STEM LEARNING SHEET · AGES 8–12

Rockets: Action, Reaction, Liftoff!

THE SCIENCE

Rockets work because of Newton's third law of motion: for every action, there is an equal and opposite reaction. When a rocket pushes gas or water out of its bottom very fast, that gas pushes the rocket up just as hard. That upward push is called thrust.

Two forces fight back. Gravity pulls the rocket down, and drag, which is air pushing against it, slows it down. A rocket rises when thrust is bigger than gravity. Fins help keep it flying straight, and a pointed nose cone helps it slice through the air with less drag.

KEY WORDS

Force: a push or a pull.

Thrust: the force that pushes a rocket forward.

Gravity: the force that pulls things toward Earth.

Drag: air pushing against a moving object.

Newton's third law: every action has an equal and opposite reaction.

Stability: staying pointed in a steady direction.

EXPERIMENT – FIN TEST

Launch your rocket 3 times with 3 fins, 3 times with 4 fins, and 3 times with no fins, keeping everything else the same. Have a partner estimate height or measure the distance it lands from the launch pad.

Think about it: How did fins change how straight it flew?

SAFETY

Launch outdoors in an open area, with adult supervision. Never stand over the rocket or point it at people, animals, buildings, or power lines. Wait until it fully lands before picking it up. Wear safety glasses.

CHECK YOUR UNDERSTANDING

1. What is thrust?
2. Finish Newton's third law: for every action...
3. What two forces slow a rocket down?
4. What do fins do?
5. Why is a nose cone pointed?

Name: _____

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ANSWER KEY (FOR GROWN-UPS)

1. The force that pushes a rocket forward.
2. there is an equal and opposite reaction
3. Gravity and drag.
4. Keep the rocket flying straight.
5. To reduce drag.

Name: _____